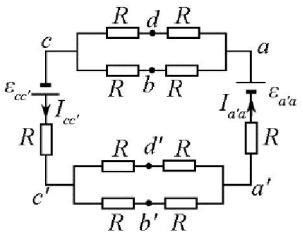
$$U_{a'a} = U_{cc'} = 0 \qquad (5b)$$

After connecting $a'a$ and cc', the currents in $a'a$ and cc' are

$$I_{aa'}^{(1)} = I_{cc'}^{(1)} = 0 \qquad (6b)$$

Thus connecting $a'a$, cc' leaves the currents in ba, ad unchanged.

Now only $\varepsilon_{a'a}$, $\varepsilon_{cc'}$ act; $\varepsilon_{b'b}$, $\varepsilon dd'$ are shorted. Find each branch current. Opening $b'b$, dd' gives the circuit shown. As above, with $\varepsilon_{a'a}$, $\varepsilon_{cc'}$ alone, currents in $a'a$, cc' are



$$I_{a'a}^{(2)} = I_{cc'}^{(2)} = \frac{\sqrt{2}}{4}\frac{Bl\omega^2}{R}\sin\theta \qquad (7b)$$

The currents in ba、 ad、 bc and cd are

$$I_{ba}^{(2)} = I_{cd}^{(2)} = -\frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}\sin\theta \qquad (8b)$$

$$I_{ad}^{(2)} = I_{bc}^{(2)} = \frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}\sin\theta \qquad (9b)$$

Across b'、 b and d、 d', $U_{b'b} = U_{dd'} = 0$. On reconnecting $b'b$, dd', currents in $b'b$ and dd' are

$$I_{b'b}^{(2)} = I_{dd'}^{(2)} = 0 \qquad (10b)$$

Thus connecting bb', dd' leaves all branch currents unchanged.

By superposition, the currents in ba、 ad、 bc and cd are

$$I_{ba} = I_{ba}^{(1)} + I_{ba}^{(2)} = \frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}\left(\cos\theta - \sin\theta\right) \qquad (11b)$$

$$I_{ad} = I_{ad}^{(1)} + I_{ad}^{(2)} = \frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}\left(\cos\theta + \sin\theta\right) \qquad (12b)$$

$$I_{bc} = I_{bc}^{(1)} + I_{bc}^{(2)} = \frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}\left(\cos\theta + \sin\theta\right) \qquad (13b)$$

$$I_{cd} = I_{cd}^{(1)} + I_{cd}^{(2)} = \frac{\sqrt{2}}{8}\frac{Bl\omega^2}{R}\left(\cos\theta - \sin\theta\right) \qquad (14b)$$

2. As in Method 1.

Scoring: use the criteria for Method 1.

VIII.

Solution:

1. In the ray diagram below, the mirror-reflected rays meet at the lens focus.